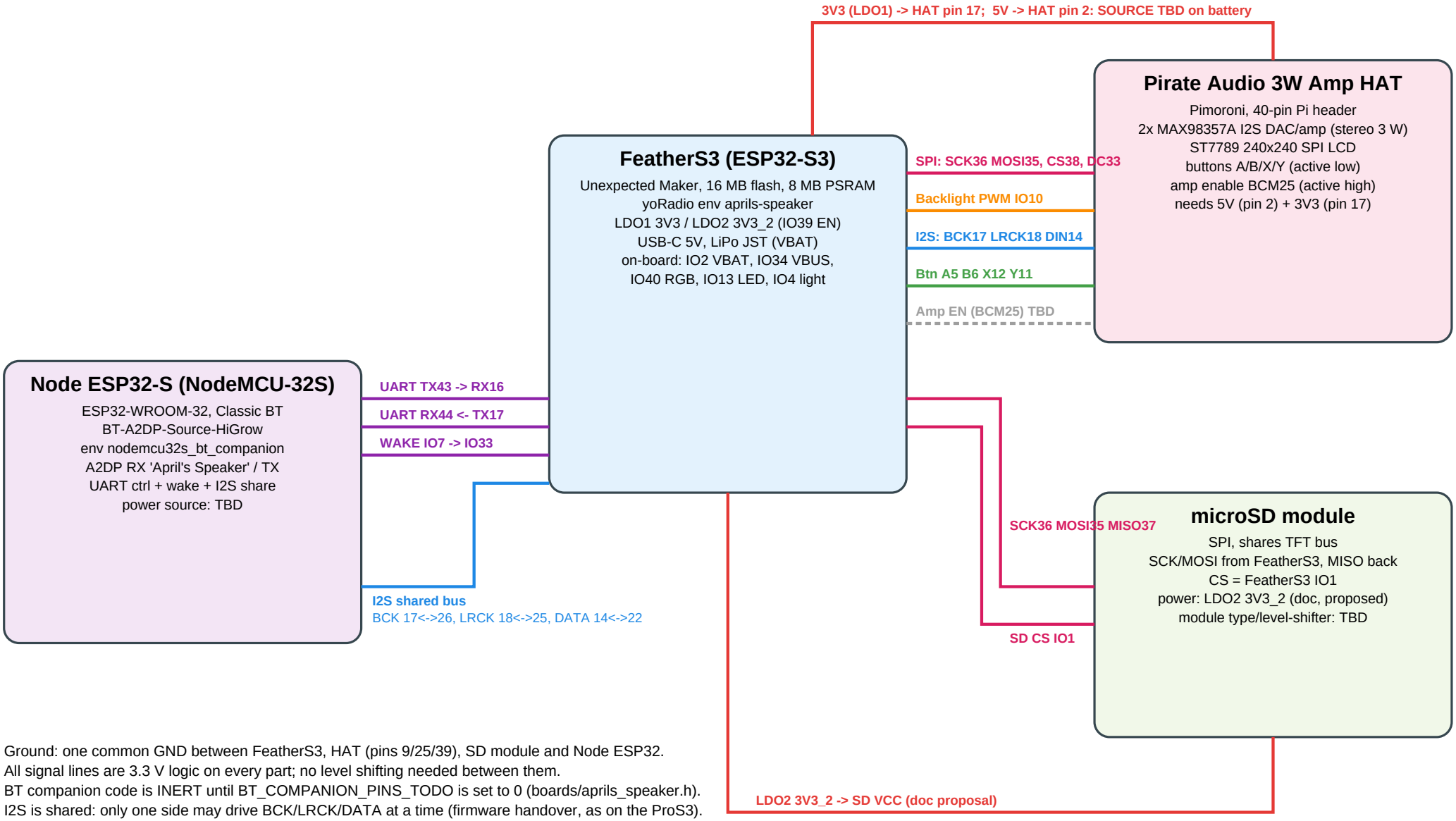
April's Speaker - overall block wiring diagram

SPI (TFT/SD) · I2S audio · Buttons · BT companion · Power · Ground · Control · On-board fixed · TBD / not assigned

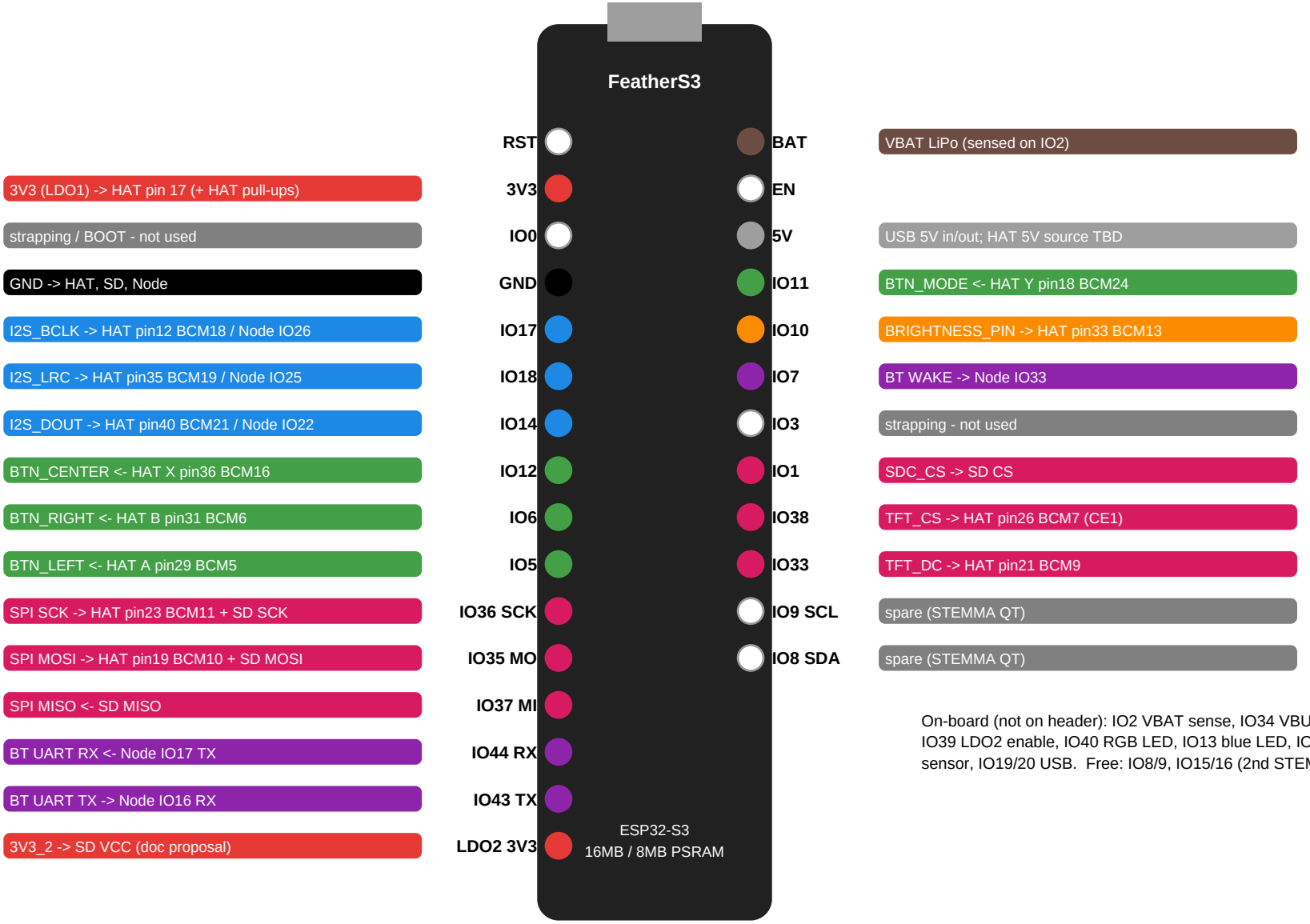


Ground: one common GND between FeatherS3, HAT (pins 9/25/39), SD module and Node ESP32.
All signal lines are 3.3 V logic on every part; no level shifting needed between them.
BT companion code is INERT until BT_COMPANION_PINS_TODO is set to 0 (boards/aprils_speaker.h).
I2S is shared: only one side may drive BCK/LRCK/DATA at a time (firmware handover, as on the ProS3).

Source: yoRadio-Fusion-ProS3 branch cursor/aprils-speaker-oct23 (boards/aprils_speaker.h, platformio.ini env aprils-speaker)
+ BT-A2DP-Source-HiGrow branch cursor/aprils-speaker-oct23 (include/options.h, env nodemcu32s_bt_companion). Pins are PROPOSED in code (PCB not final).

April's Speaker - FeatherS3 pinout (top view, USB-C at top)

SPI (TFT/SD) | I2S audio | Buttons | BT companion | Power | Ground | Control | On-board fixed | TBD / not assigned



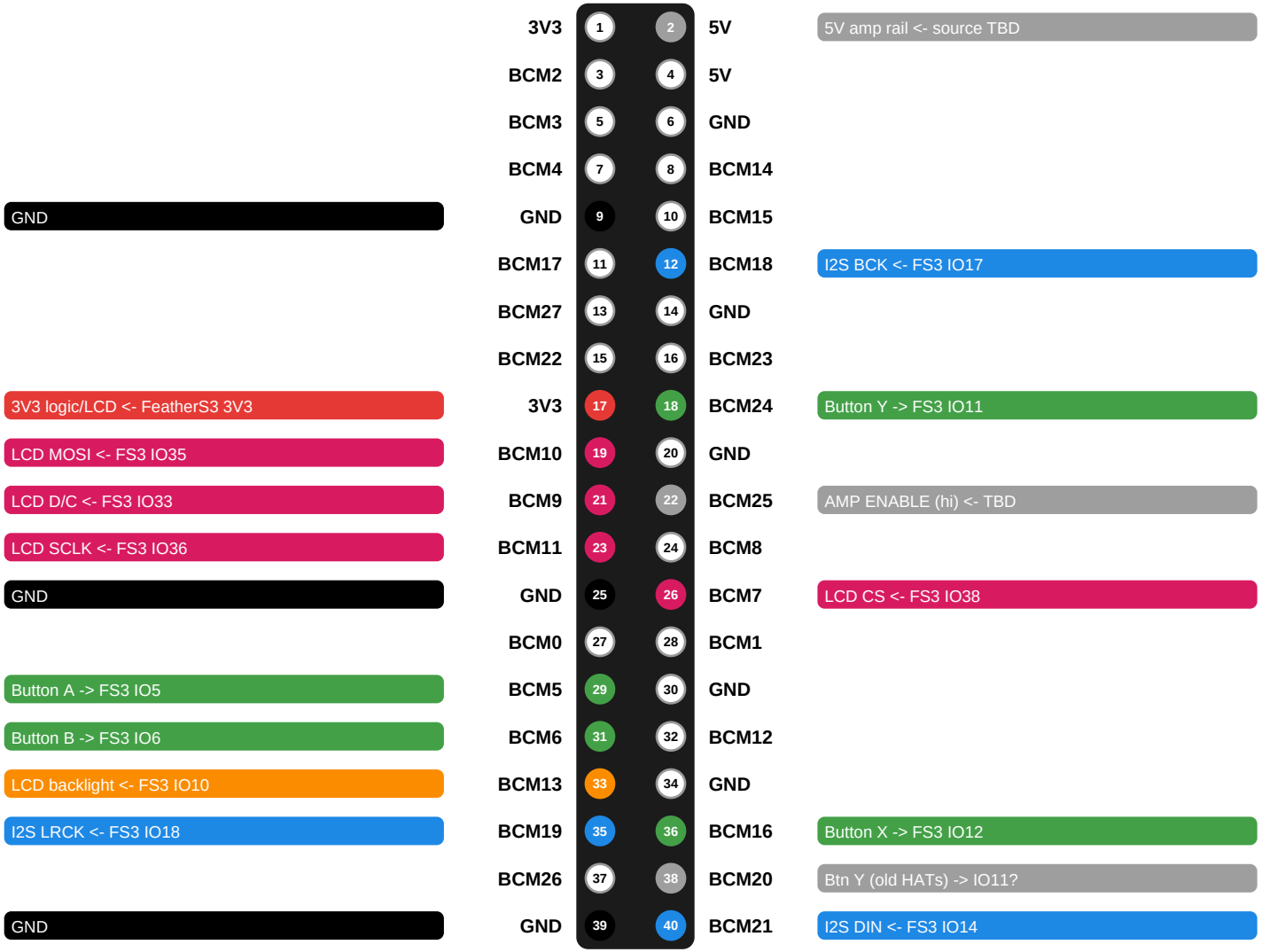
On-board (not on header): IO2 VBAT sense, IO34 VBUS sense, IO39 LDO2 enable, IO40 RGB LED, IO13 blue LED, IO4 light sensor, IO19/20 USB. Free: IO8/9, IO15/16 (2nd STEMMA).

Physical order verified against Unexpected Maker 'FeatherS3_Pin_Reference.png' and kicad/symbols/FeatherS3.kicad_sym (github.com/UnexpectedMaker/esp32s3).
Firmware defines: boards/aprils_speaker.h. TFT_RST = -1 (HAT has no reset line).

Source: yoRadio-Fusion-ProS3 branch cursor/aprils-speaker-oct23 (boards/aprils_speaker.h, platformio.ini env aprils-speaker)
+ BT-A2DP-Source-HiGrow branch cursor/aprils-speaker-oct23 (include/options.h, env nodemcu32s_bt_companion). Pins are PROPOSED in code (PCB not final).

April's Speaker - Pirate Audio 3W Amp HAT - 40-pin header (Pi numbering, pin 1 top-left)

SPI (TFT/SD) | I2S audio | Buttons | BT companion | Power | Ground | Control | On-board fixed | TBD / not assigned

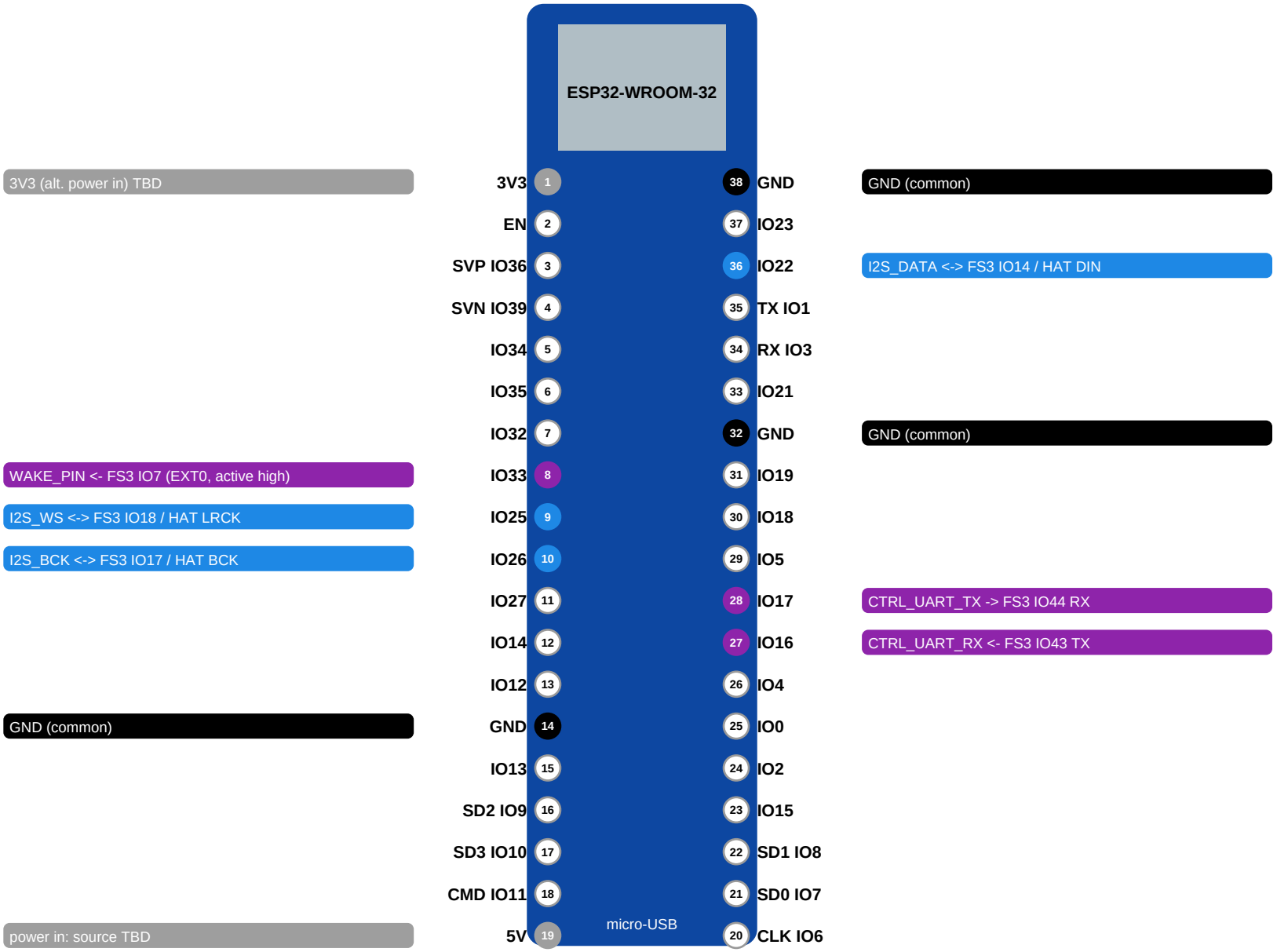


Pin roles verified against pinout.xyz 'pimoroni-pirate-audio-3w-stereo-amp' (power 2,17; ground 9,25,39; I2S 12/35/40; amp enable 22 active high; buttons 29/31/36/18 (old boards Y = 38); LCD 33 backlight, 21 D/C, 19 MOSI, 23 SCLK, 26 CS) and the Pimoroni product page (2x MAX98357A, ST7789 240x240, buttons active low BCM 5/6/16/24, 'DAC enable pin BCM 25 must be driven high'). Unused pins are white.

Source: yoRadio-Fusion-ProS3 branch cursor/aprils-speaker-oct23 (boards/aprils_speaker.h, platformio.ini env aprils-speaker)
+ BT-A2DP-Source-HiGrow branch cursor/aprils-speaker-oct23 (include/options.h, env nodemcu32s_bt_companion). Pins are PROPOSED in code (PCB not final).

April's Speaker - Node ESP32-S (NodeMCU-32S, 38-pin) - top view, antenna at top

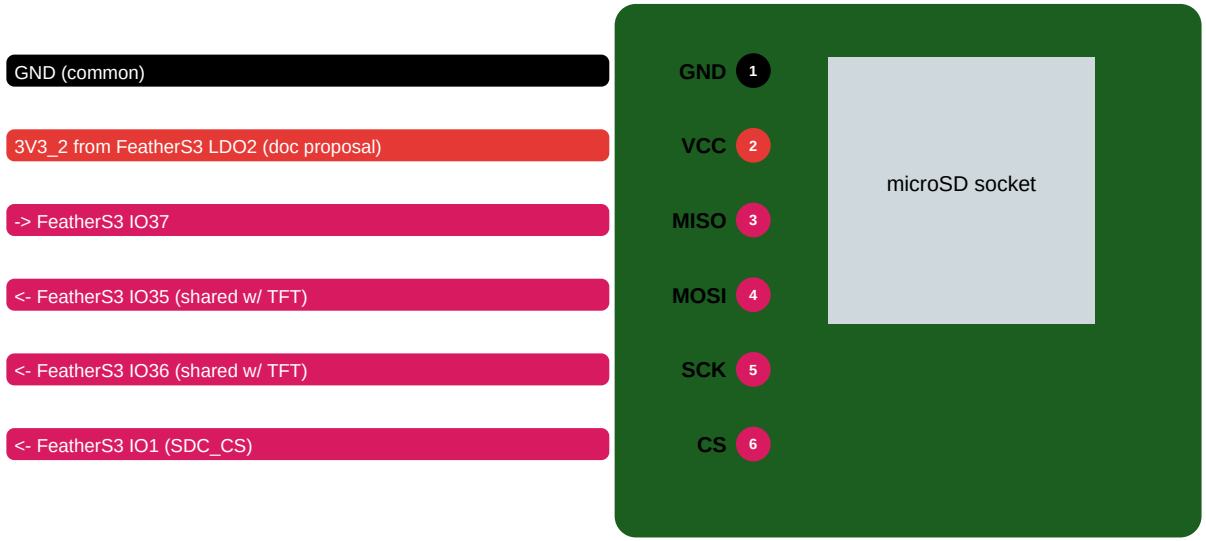
SPI (TFT/SD) | I2S audio | Buttons | BT companion | Power | Ground | Control | On-board fixed | TBD / not assigned



Physical order verified against the Ai-Thinker 'NodeMCU-32S specification v1.3' pin table (pins 1-19 left, 20-38 right from the USB end). Firmware: include/options.h COMPANION_BOARD_NODEMCU32S block. Avoided: flash pins 6-11, straps 0/2/5/12/15, UART0 1/3. A 30-pin 'ESP32 DevKit V1' has a different order - check which board April has.

Source: yoRadio-Fusion-ProS3 branch cursor/aprils-speaker-oct23 (boards/aprils_speaker.h, platformio.ini env aprils-speaker)
+ BT-A2DP-Source-HiGrow branch cursor/aprils-speaker-oct23 (include/options.h, env nodemcu32s_bt_companion). Pins are PROPOSED in code (PCB not final).

SPI (TFT/SD) I2S audio Buttons BT companion Power Ground Control On-board fixed TBD / not assigned



The code fixes only the signals (SDC_CS 1, SPI 36/35/37, SDSPISPEED 10 MHz, SD_SHARE_TFT_SPI 1); it does not name a module, so the PHYSICAL pin ORDER above is a common 6-pin layout and must be checked against the silkscreen of April's actual module (TBD). Use a 3.3 V module (or one whose regulator/level shifter accepts 3.3 V); 5 V-only AMS1117 + 74LVC125 modules need VCC = 5 V. Recommended: ~100 R series resistors on SD lines and a 10 k pull-up on CS (ProS3 practice). Firmware drives SDC_CS and TFT_CS HIGH at boot before SPI starts (SD park) and powers LDO2 (IO39) first.

Source: yoRadio-Fusion-ProS3 branch cursor/aprils-speaker-oct23 (boards/aprils_speaker.h, platformio.ini env aprils-speaker)
+ BT-A2DP-Source-HiGrow branch cursor/aprils-speaker-oct23 (include/options.h, env nodemcu32s_bt_companion). Pins are PROPOSED in code (PCB not final).

April's Speaker - master connection table

#	From (part / pin)	To (part / pin)	Signal	Notes
1	FeatherS3 IO36 (SCK)	HAT pin 23 (BCM11) + SD SCK	SPI SCLK	shared TFT/SD bus, TFT 20 MHz / SD 10 MHz; 3.3 V
2	FeatherS3 IO35 (MO)	HAT pin 19 (BCM10) + SD MOSI	SPI MOSI	shared
3	SD MISO	FeatherS3 IO37 (MI)	SPI MISO	HAT does not use MISO
4	FeatherS3 IO38	HAT pin 26 (BCM7, CE1)	TFT_CS	driven HIGH at boot
5	FeatherS3 IO33	HAT pin 21 (BCM9)	TFT_DC	
6	-	HAT (LCD reset)	TFT_RST	none: TFT_RST -1, HAT has no reset line
7	FeatherS3 IO10	HAT pin 33 (BCM13)	Backlight PWM	BRIGHTNESS_PIN (LEDC)
8	FeatherS3 IO1	SD CS	SDC_CS	driven HIGH at boot (SD park); 10 k pull-up recommended
9	FeatherS3 IO17	HAT pin 12 (BCM18) + Node IO26	I2S BCLK	shared with companion (one driver at a time)
10	FeatherS3 IO18	HAT pin 35 (BCM19) + Node IO25	I2S LRCK/WS	
11	FeatherS3 IO14	HAT pin 40 (BCM21) + Node IO22	I2S DATA	no MCLK (MAX98357A needs none)
12	TBD / not assigned	HAT pin 22 (BCM25)	AMP ENABLE	active HIGH per Pimoroni; no pin in code (MUTE_PIN 255) - see open questions
13	HAT pin 29 (BCM5) Btn A	FeatherS3 IO5	BTN_LEFT	active low; HAT pull-ups + internal pull-up; RTC GPIO
14	HAT pin 31 (BCM6) Btn B	FeatherS3 IO6	BTN_RIGHT	active low

Source: yoRadio-Fusion-ProS3 branch cursor/aprils-speaker-oct23 (boards/aprils_speaker.h, platformio.ini env aprils-speaker)
+ BT-A2DP-Source-HiGrow branch cursor/aprils-speaker-oct23 (include/options.h, env nodemcu32s_bt_companion). Pins are PROPOSED in code (PCB not final).

#	From (part / pin)	To (part / pin)	Signal	Notes
15	HAT pin 36 (BCM16) Btn X	FeatherS3 IO12	BTN_CENTER	active low
16	HAT pin 18 (BCM24) Btn Y	FeatherS3 IO11	BTN_MODE	active low; deep-sleep wake (WAKE_PIN1). Old HATs: pin 38 (BCM20)
17	FeatherS3 IO43 (TX)	Node IO16	BT UART TX->RX	115200; ROM boot log also appears here at reset
18	Node IO17	FeatherS3 IO44 (RX)	BT UART RX<-TX	115200
19	FeatherS3 IO7	Node IO33	BT WAKE	EXT0 active high, 600 ms pulse, pulled down on Node
20	FeatherS3 3V3 (LDO1)	HAT pin 17	3V3	LCD logic + button pull-ups; LDO1 stays on in deep sleep
21	FeatherS3 LDO2 3V3_2	SD VCC	3V3_2	doc proposal (not in code); off in deep sleep; LDO2 EN = IO39
22	5 V source TBD	HAT pin 2	5V amp rail	FeatherS3 5V pin is USB only; on battery needs boost - see open questions
23	Power source TBD	Node 5V (pin 19) or 3V3 (pin 1)	Node power	not specified anywhere
24	FeatherS3 GND	HAT pins 9, 25, 39; SD GND; Node GND (pins 14/32/38)	GND	single common ground
-	FeatherS3 on-board	IO2 VBAT, IO34 VBUS, IO39 LDO2 EN, IO40 RGB, IO13 LED, IO4 light	fixed	not wired externally

Source: yoRadio-Fusion-ProS3 branch cursor/aprils-speaker-oct23 (boards/aprils_speaker.h, platformio.ini env aprils-speaker)
+ BT-A2DP-Source-HiGrow branch cursor/aprils-speaker-oct23 (include/options.h, env nodemcu32s_bt_companion). Pins are PROPOSED in code (PCB not final).

POWER & GROUND
- FeatherS3: USB-C 5 V or 1S LiPo (JST, max 4.2 V). LDO1 3V3 (700 mA) always on; LDO2 3V3_2 (700 mA) enabled by IO39, off in deep sleep, also powers the RGB LED.
- HAT 3V3 (pin 17) from LDO1 so the button pull-ups stay valid in deep sleep (Y is the wake button). Blank panel with backlight on = pin 17 not connected.
- HAT 5V (pin 2) feeds the 2x MAX98357A amps. The FeatherS3 '5V' pin carries USB 5 V only - on battery the amp needs a boost converter or another source (TBD).
- SD module on LDO2 (doc proposal) keeps card inrush off the MCU rail; ~100 R series resistors on SD lines recommended.
- One common ground for all four parts. All logic is 3.3 V; MAX98357A inputs accept 3.3 V logic at 5 V supply.

OPEN QUESTIONS / TBD (not defined in code - nothing invented)
1. Amp enable HAT pin 22 (BCM25): no GPIO assigned (MUTE_PIN 255). Pimoroni says it must be driven high - tie to 3V3, or assign a spare GPIO (e.g. IO15/16/8/9)?
2. 5 V source for HAT pin 2 on battery (boost converter? which?).
3. Node ESP32 power: 5 V pin or 3V3 pin, and from which rail?
4. All BT companion pins are PROPOSED (BT_COMPANION_PINS_TODO 1, 'pins are a proposal until wiring is confirmed') - awaiting Cain's wiring answers.
5. Which Node board exactly: 38-pin NodeMCU-32S (env board nodemcu-32s, drawn here) or a 30-pin DevKit V1 (different physical order)?
6. microSD module type: physical pin order, 3.3 V vs 5 V module, CS pull-up fitted?
7. Button Y: new HAT (pin 18, BCM24) vs old HAT (pin 38, BCM20) - check April's board revision.
8. Code comments said 'PCM5102A DAC' - the 3W Amp actually uses 2x MAX98357A (the PCM5100A is on the Line-Out HAT). Comments corrected; pins unaffected.
9. Whole FeatherS3 pin map is PROPOSED (main PCB not final) and never bench-tested.

SOURCES
- Unexpected Maker FeatherS3: https://esp32s3.com/feathers3.html ; pin card + KiCad symbol: https://github.com/UnexpectedMaker/esp32s3 (Pinout Cards/FeatherS3_Pin_Reference.png, kicad/symbols/FeatherS3.kicad_sym)
- Pimoroni Pirate Audio 3W Stereo Amp: https://shop.pimoroni.com/products/pirate-audio-3w-stereo-amp ; https://github.com/pimoroni/pirate-audio
- pinout.xyz: https://pinout.xyz/pinout/pirate_audio_3w_amp (source src/en/overlay/pimoroni-pirate-audio-3w-stereo-amp.md)
- Ai-Thinker NodeMCU-32S specification v1.3: https://component.aithinker.com/_media/nodemcu32-s_specification_v1.3.pdf
- Firmware: yoRadio-Fusion-ProS3 boards/aprils_speaker.h, docs/APRILS_SPEAKER.md; BT-A2DP-Source-HiGrow include/options.h

Source: yoRadio-Fusion-ProS3 branch cursor/aprils-speaker-oct23 (boards/aprils_speaker.h, platformio.ini env aprils-speaker)
+ BT-A2DP-Source-HiGrow branch cursor/aprils-speaker-oct23 (include/options.h, env nodemcu32s_bt_companion). Pins are PROPOSED in code (PCB not final).